

Advanced (Habanero)

- Q1.** What is the vertex of the parabola $y = x^2 + 4x + 4$?
- (A) (0, 0)
(B) (0, 4)
(C) (-2, 0)
(D) (-2, -4)
(E) (2, 4)
- Q2.** Which equation represents a parabola with axis of symmetry $x = -3$?
- (A) $y = x^2 + 6x + 9$
(B) $y = x^2 - 6x + 9$
(C) $y = x^2 + 3x - 2$
(D) $y = x^2 - 3x - 2$
(E) $y = x^2 + 2x + 1$
- Q3.** What is the y -intercept of the parabola $y = x^2 - 4x - 3$?
- (A) -3
(B) -2
(C) -1
(D) 0
(E) 3
- Q4.** What is the minimum value of the parabola $y = x^2 + 2x + 1$?
- (A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q5.** Which equation represents a parabola with a maximum value of -2 ?
- (A) $y = -x^2 + 2$
(B) $y = -x^2 - 2$
(C) $y = x^2 + 2$
(D) $y = x^2 - 2$
(E) $y = -x^2 + 4$
- Q6.** What is the axis of symmetry of the parabola $y = x^2 - 2x - 3$?
- (A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $x = 2$
(E) $x = 3$
- Q7.** What is the direction of the parabola $y = -x^2 + 2$?
- (A) Up
(B) Down
(C) Left
(D) Right
(E) None
- Q8.** What is the x -intercept of the parabola $y = x^2 + 2x + 1$?
- (A) -1
(B) -2
(C) 0
(D) 1
(E) 2

Open Ended Questions (FRQ)

Q9. Graph the parabola $y = x^2 - 4x + 3$ and identify its vertex, axis of symmetry, and y -intercept.

Q10. Find the maximum or minimum value of the parabola $y = -x^2 + 2x + 1$ and explain your reasoning.

Chef's Tips

- Provide a clear and concise prompt for each question.
- Ensure that each question assesses a specific learning objective.
- Make sure students have enough space to answer each question.